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Review

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Light pollution: A review of the scientific literature

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Jesús Rodrigo-Comino,[1] Stephan Seeling,[2] Manuel K. Seeger[2] and Johannes B. Ries[2]

Abstract

Light pollution is the consequence of elevated lighting emitted by human-made artefacts to the lower atmosphere. Recently, there have been major advances in the assessment and mitigation of light pollution impacts on humans and the natural ecosystems. Severe negative impacts of light pollution have been highlighted while very few mitigation measures have been implemented. People (scientists, policymakers or stakeholders) interested in light pollution may not have a holistic perspective of the problem, and also there is a need for incorporating social and natural dimensions. Therefore, the main goal of this paper is to review the literature on light pollution using ISI Web of Science by paying attention to the (i) type of publication, year and journal; (ii) impacts on specific elements; (iii) location and (iv) methods used. Our results indicated that the elevated number of papers come from a diverse range of disciplines, methods, places and scales. It is clear that light pollution is getting enough attention from the scientific community but decisions on the implementation of mitigation measures are left with the stakeholders, ordinary inhabitants, policymakers and politicians. Nevertheless, light pollution is having impacts on the health of humans and the natural ecosystem as perceived by experts and inhabitants having divergent perspectives. Thus, light pollution is multifaceted but difficult to be faced, mitigated and not holistically understood. This review paper groups the total impacts of light pollution on the Earth presents some contradictory results, summarises mitigation measures, and provides specific future research directions.

Keywords

artificial lights, human health, light pollution, natural ecosystem health

[1]University of Granada, Spain
[2]Trier University, Germany

Corresponding author:
Jesús Rodrigo-Comino, Departamento de Análisis Geográfico Regional y Geografía Física, Facultad de Filosofía y Letras, Campus Universitario de Cartuja, University of Granada, Granada 18071, Spain.
Email: jesusrc@ugr.es

Introduction

An increase in human activities during the last few centuries (19th, 20th and 21st) are having negative impacts on the natural ecosystems, and humankind lifestyles and health (Rodrigo-Comino et al., 2020; Trenberth, 2018; Weber, 2003). One of the most important human activities is the increase in land consumption (Panagos and Katsoyiannis, 2019; Salvati and Carlucci, 2016) and the lack of sustainable and efficient land management plans (Cohen-Shacham et al., 2016; De Groot et al., 2010; De Sainte Marie, 2014). As a consequence, humankind will have to face adverse land degradation processes and difficult challenges related to their impacts such as global warming (Dai, 2011; IPCC, 2018), water pollution (Kumar et al., 2019), loss of soil fertility and productivity (Pieri, 1992; Wu et al., 2011), diseases such as the COVID-19 (Reuschke and Felstead, 2020) and light pollution (Longcore and Rich, 2004).

Light is a key resource for the activation of some vital processes such as photosynthesis, the partition of daily and night activities or help to recover and repair organisms, and conservation of light environments. Also, it informs the circadian clocks and photoperiodism, helping visual perception and spatial orientation (Gaston et al., 2013). Artificial lights used to illuminate streets, homes and buildings emit light to the lower atmosphere that remains there due to the high concentration of aerosol particles, and continuous emission can be dangerous (Ściężor and Czaplicka, 2020). Bugnot et al. (2019) stated that light pollution is normally assessed at local scales, but there is a mismatch between 'the spatial extent of the impacts and that of the pressures driving those impacts'. Light pollution is defined as a 'multi-facet' term (Luginbuhl et al., 2010), and to date, there is no a clear understanding of its impacts on human health (Cho et al., 2015; Rybnikova et al., 2016) and the natural ecosystems (Grubisic et al., 2018b; Guetté et al., 2018). The concentration of elevated amounts of electromagnetic energy during night-time from the Earth's surface light sources in specific territories are having negative impacts on astronomical observations, and also on living organisms, including humans (Kocifaj, 2016). Kyba et al. (2017) estimated that artificial light transferred to the atmosphere could reach 2.2% per year, with considerable differences among countries and hemispheres, but will remain stable or decrease in a few areas. Their results are not consistent with other assessments made to estimate energy reductions at a global scale but highlighted the increasing trend in light pollution. Following Kyba's (2018) recommendations, people (inhabitants, stakeholders and policymakers) awareness would be vital to modify lighting technologies and emissions. Macgregor et al. (2017) stated that humankind requires 'suitable lighting technologies and concepts which are ecologically, socially and economically sustainable'. They insist that managing darkness is an essential part of future conservation and lighting management plans of each modern society over the world is imperative. However, without a global consensus, unpredictable environmental and sociological outcomes could appear. This cannot be reached without a complete understanding of this multifaceted problem. Considering the worldwide population growth, the elevated concentration of inhabitants in urban areas and the corresponding infrastructures and services (industries, roads, airports, railways, etc.), light pollution is an environmental issue showing an exponentially increasing trend that is difficult to control. As a result, the scientific community have been paying attention to this issue during the last two decades.

The scientific literature on light pollution is diverse, focussing on different impacts of light pollution (animals, human well-being, etc.), using distinct methods or approaches (e.g. remote sensing techniques, surveys, laboratory analysis, modelling tools, etc.) and considering different spatiotemporal scales (e.g. experimental plots, urban or rural areas, regions, country). This shows the great advances conducted to assess, reduce or avoid this environmental problem. However, we consider that the reader (scientists, policymakers or stakeholders) could lose the general perspective of the problem if their expectations to get an overall idea of the problem is not satisfied.

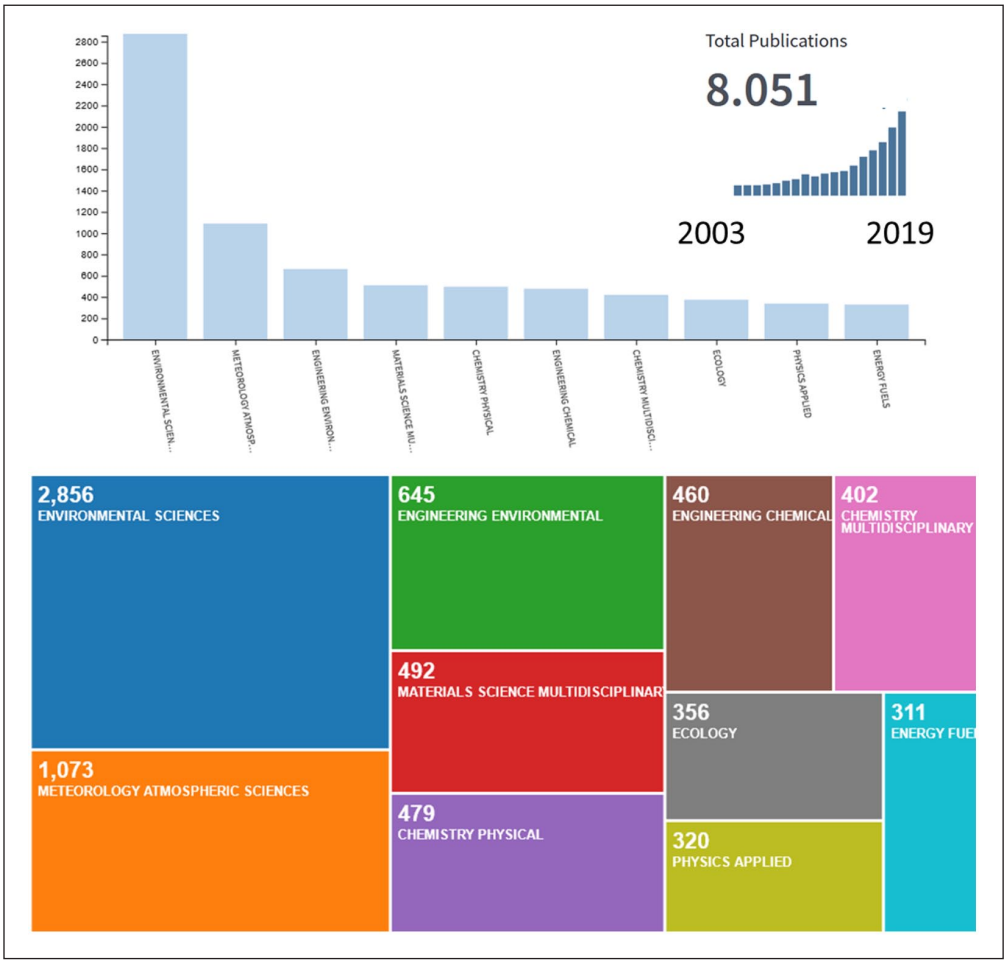


Figure 1. Results from ISI Web of Science investigations conducted from 2003 to 2019 that contains the term 'light pollution'.

Conducting a simple search using the ISI Web of Science to find literature up to 2019 that contain the terms 'light pollution', >8000 articles were found (Figure 1). Obviously, not all of the articles aim to investigate light pollution. Therefore, the main goal of this review paper was to assess the whole available literature related to the term 'light pollution', paying attention to the (i) type of publications, year and journal; (ii) impacts on specific elements; (iii) location and (iv) methods used. We hypothesise that the number of specific papers aimed to investigate light pollution is high and more diverse (coming from different disciplines, methods, places and different scales). Besides, we believe that this review will be able to group the total impacts of light pollution on the Earth, showing some contradictory results, summarising solutions and offering future research directions.

Materials and methods

To achieve the above-mentioned goal, we searched the term 'light pollution' in the ISI Web of Science (https://apps.webofknowledge.com/WOS_GeneralSearch_input.do?product=WOS&search_mode=

GeneralSearch&SID=F5ZJLB669fH5Bx5kkcR&preferencesSaved=) from the beginning of the database (2003) to 2019. A total of 8051 publications were obtained on October 3, 2020. It needs to be pointed out the authors of this review paper collectively can work in English, German, Spanish and French languages. However, we restricted our search on publications written in English only for the general scientific community, considering English as a vehicular language (Clouet, 2017; Gordin, 2015). During the literature selection, all the abstracts were assessed to select the only ones containing the information about light pollution in the introduction, methods, results or conclusions. Books were not included and only one PhD thesis, indexed in this database, was taken into account. In the end, 621 publications were used. Therefore, there may be valuable information published in other literature finders (e.g. Google Scholar), other different publication formats or distinct languages that were not considered.

The information was organised and classified in an Excel datasheet under the following criteria: (i) journals; (ii) year of publication; (iii) type of format (considering different editorial and journal formats); (iv) impacts on specific Earth's species or elements (animals, humans, light pollution in general and urban or rural areas); (v) location (place and country) and (vi) methods used.

Some issues were found during the literature search (June 2019–October 2020) of the systematic review considering Haddaway et al. (2020). Regarding the division of the impacts on specific elements, we paid attention to the main topic highlighted in the title, abstract and the main conclusions. In fact, some of the articles could be classified into different groups due to the multidisciplinary nature of the topic. The study area or the country, where the investigation was conducted were difficult to find in some manuscripts. Therefore, if the specific study area or laboratory was not mentioned in the text, we tagged that manuscript as 'non-classified'. For some of them, we took the place of the laboratory where the assessment or models were applied. In the case of the remote sensing applications, the location of the investigation was used as the study area. The complete list with all the titles used in this review will be posted on the website of the Smart Light-HUB (http://smartlighthub.eu/), which is an international project (INTERREG) related to the co-creation, diffusion and transfer of knowledge on light pollution, that funded this study. Specific details will be shared upon request.

Results and discussion

Publication trends and journals

A total of 621 publications were selected in this review. As above-mentioned, these articles mainly investigate light pollution and discussed its impacts on the specific sphere. Figure 2 indicates that only 10 articles were published up to 2005, and 7 were recorded in 2006. However, a linear increase was registered ($R^2 = 0.87$) from 2007 to 2019, reaching >100 papers in 2019. These results show the increasing importance of this topic for the scientific community just like other important environmental issues such as urban climate change (e.g. Sharifi, 2021), urban sprawl, food security and agricultural systems in developing countries (e.g. Abu Hatab et al., 2019), strategic environment assessments (e.g. Caschili et al., 2014), soil erosion (e.g. Nearing et al., 2017; Rodrigo-Comino, 2018) or emission of pesticides to the air (e.g. Gil and Sinfort, 2005).

Journals with five or more publications related to light pollution are presented in Figure 3. It is worthy to highlight the diversity of the journals where the 276 papers were published. There are at least 190 scientific journals with at least one article on light pollution. The journals with the highest number of papers published on this topic in decreasing order are (i) Journal of Quantitative Spectroscopy and Radiative Transfer (32); (ii) Scientific Reports (18); (iii) Environmental Pollution (16); (iv) Journal of Experimental Zoology – A (16) and (v) Lighting Research & Technology (15).

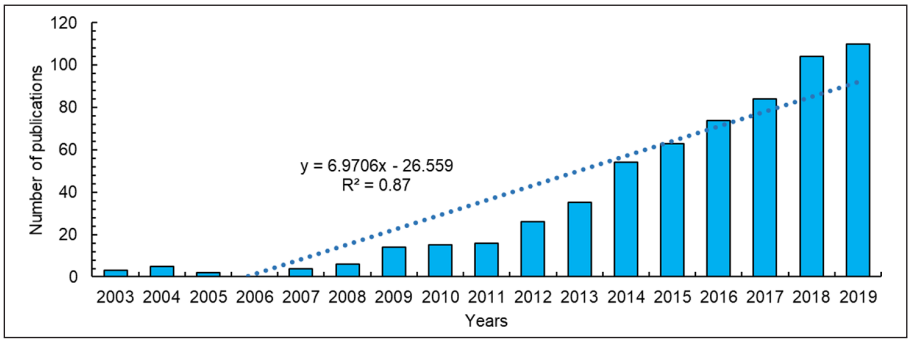


Figure 2. Number of scientific publications and recent trend on light pollution based on ISI Web of Science from 2003 to 2019.

The scope of the journals with the most articles (Figure 3) aim at astronomy (e.g. Monthly Notices of the Royal Astronomical Society or Publications of the Astronomical Society of the Pacific), miscellanea (e.g. Scientific Reports, Plos One or Science of the Total Environment), remote sensing applications and developments (e.g. Remote sensing or Remote sensing of the Environment), ecology and biology (e.g. Global Change Biology, Proceedings of the Royal Society B: Biological Sciences or Journal of Applied Ecology) and engineering (e.g. Lighting Research & Technology, Applied Optics or Sensors). The ensuing sections demonstrate whether the increase in the number of publications and diversity of journals is linked to an 'increase of actionable knowledge' (Bornmann and Mutz, 2015). This is the key to develop efficient solutions and sustainable management plans.

Global distribution of publications

Figure 4 is depicted the number of publications on light pollution per country between 2003 and 2019, and Table 1 notes the year of publication per country. It is important to mention that it was not possible to classify 134 articles because there was no information on the study area, laboratory or any specific place. Here, we also included review papers, editorials, letters to editors, etc. Twenty-three articles considered two or more study areas from different countries. A total of 57 different countries were found in studies assessing light pollution from different points of view. Research groups from Chile, Italy and the USA published the first pioneering articles indexed in ISI Web of Knowledge in 2003.

The country with the highest number of publications is the USA (91), which experienced an increase in research from 9 in 2013 to 18 articles per year in 2019. Several studies were conducted in Columbus under laboratory conditions (e.g. Cissé et al., 2017; Sullivan et al., 2019) or in big cities such as New York, Houston, Los Angeles or Las Vegas (e.g. Cabrera-Cruz et al., 2019; Duriscoe et al., 2018). Germany (40) followed the USA with an average of ≈4 articles per year of publications since 2013, with peaks in 2014 (6) and 2018 (9). There are several studies conducted in Berlin (e.g. Jechow et al., 2019; Kyba et al., 2011), Munich (e.g. Caorsi et al., 2019) or Potsdam (e.g. Puschnig et al., 2014) that assessed the problems related to urban areas. Also, it is relevant to remark on the research plans performed at the Westhavelland Nature Park (e.g. Grubisic et al., 2018a; Manfrin et al., 2018). Australia also registered an elevated number of publications with more than five papers per year from 2015 in Melbourne (e.g. Willmott et al., 2019), Sidney (e.g. Haddock et al., 2019) and Brisbane (e.g. Levin et al., 2014), mostly focussed

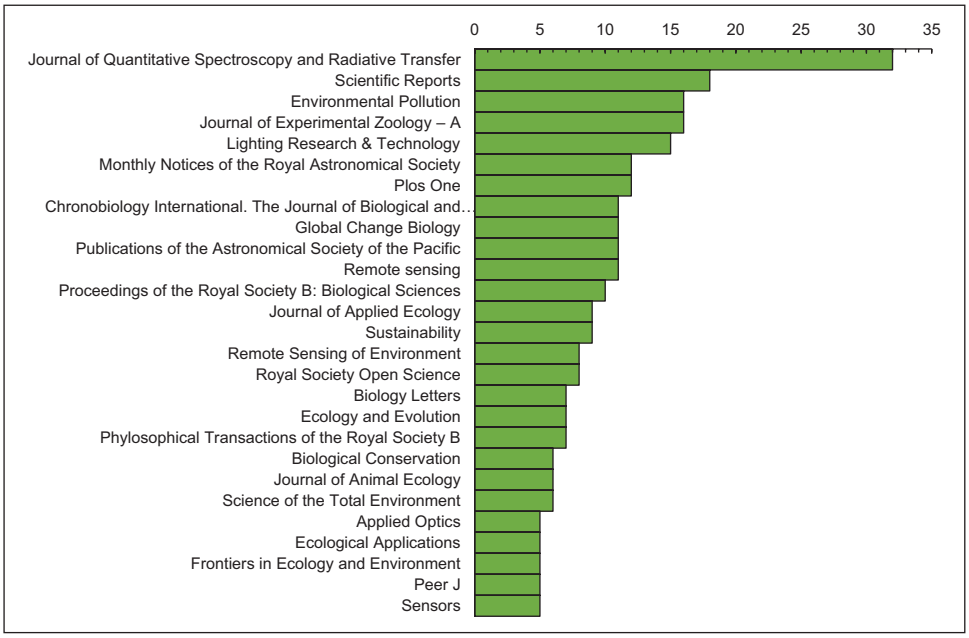


Figure 3. Name of journals where research on light pollution have been published (2003–2019). Data were sourced from ISI Web of Science.

on urban animals and coastal/island aquatic ecosystems (e.g. Pendoley and Kamrowski, 2016; Rodríguez et al., 2018). These above-mentioned countries are followed by Spain (28), The Netherlands (28), United Kingdom (27) and China (25) with the indicated number of studies.

We next discuss the location of the studies. It is observed that countries and regions with the most studies are characterised by intense urbanisation processes such as urban sprawl, the concentration of activities, population and development of transport infrastructures located in developed countries, especially in Europe, the USA and some cities of Australia (Bennie et al., 2014; Dunnett, 2015; Lamphar, 2020), but there are some contrasting trends among the countries and the regions naturally. Only a few studies were conducted in Africa, mainly in urban areas of particular importance considering its spatial expansion, demographic profiles, and prevalence of informal economies and settlements (Dodman et al., 2017). Another remarkable situation is the scarcity of studies in countries located in higher latitudes such as Sweden, Norway, Finland, Northern Russia or Canada, characterised by fewer sunlight hours and the necessity to use more hours of lights but of low population density. It is also interesting to note the increasing number of studies in the Middle East (Pakistan or Iraq) and southern or Eastern Asia (e.g. China, Hong Kong, South Korea, Malaysia, Indonesia). The main aim of most of these studies was to highlight the relevance of the use of urban street and advertising lights (Ngarambe and Kim, 2018), strong urbanisation process (Butt, 2012; Pun and So, 2012; Zhang and Hu, 2017) and civil conflicts (Li et al., 2018). In South America, studies conducted in Chile, Brazil, Mexico, Costa Rica and Argentina were not located in the main cities or their capital, but in natural areas considering issues related to animals (González et al., 2014; Simões et al., 2017), astronomical observations/tourism (Krisciunas et al., 2010; Plauchu-Frayn et al., 2017) and human health (Alaimo et al., 2019; Harb et al., 2015).

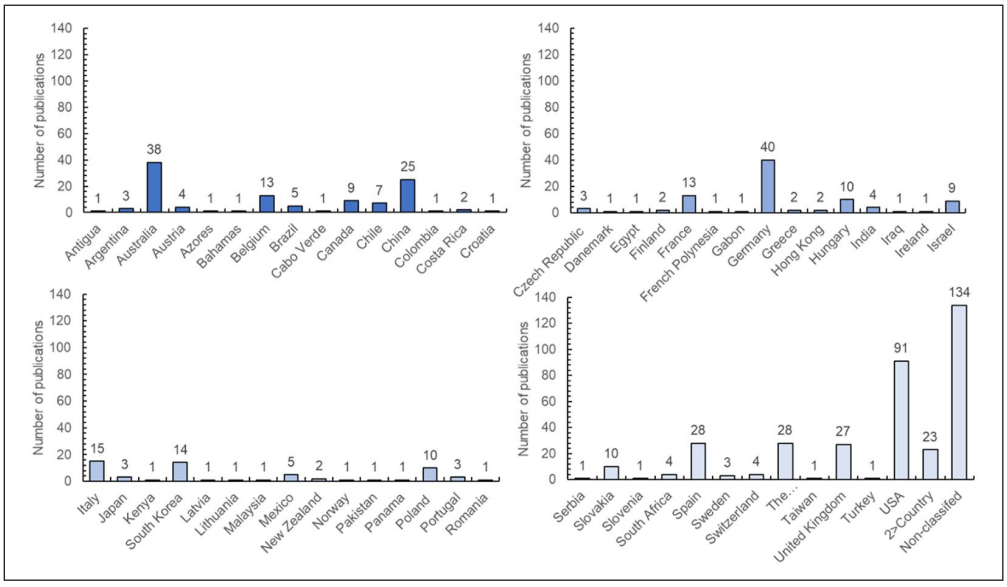


Figure 4. Number of publications on light pollution (2003–2019) per country.
Data were sourced from ISI Web of Science.

Methodologies and tools

To facilitate the understanding of the wide spectrum of methods used to assess light pollution, we first divided them into formats published by the different journals and editorials (Figure 5) and, then, on the methods applied (Figure 6).

A total of 34 different formats were found in all the surveyed journals with the format-method classification. About 80% of the articles were published under 'research article' format, followed by 9.9% under review format and the remainder falling under the other formats (e.g. editorials, brief communication, communication, short communication or scientific note).

The first group under methods is composed of articles related to sampling and surveys (35.1%). This group included techniques such as nested sampling design using light traps that catch insects or birds during different nights (Merckx and Van Dyck, 2019). Also, night-time transects and overhead flight activity surveys were commonly used for 1-day fieldwork campaigns (Macgregor et al., 2017). Non-invasive techniques through laboratory observations (Newman et al., 2015), questionnaires (Mascovich et al., 2018) and in combination with other methods such as images obtained from Google Earth (Jin et al., 2017) were also used. Despite that they are the most commonly used methods (high accuracy supposed after using in situ measurements), several articles remark the difficulties related to (i) unforeseen weather conditions; (ii) obstacles in the terrain (Ciach and Fröhlich, 2019) and (iii) the elevated time exposure and costs required to gather enough samples meaningful results (Russo et al., 2019).

The second group is characterised by remote sensing techniques (19.5%). Here, we included methods related to differential photometry with calibrated digital cameras (Jechow, 2019). Also, night-time images were taken with DSLR cameras from the International Space Station (ISS) (Sánchez de Miguel et al., 2019), Luojia 1-01 or the Suomi National Polar-Orbiting Partnership Visible Infrared Imaging Radiometer Suite (NPP-VIIRS) (Jiang et al., 2018). There are also articles related to sensors such as the Dual-frequency IDentification SONar (DIDSON) (Bolton et al.,

Table 1. Number of publications on light pollution per country and year from 2003 to 2019.

	Total	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010	2009	2008	2007	2006	2005	2004	2003
Antigua	1	1																
Argentina	3	3																
Australia	38	5	5	7	7	6	3	1	3				1					
Austria	4		1	1			1				1							
Azores	1								1									
Bahamas	1			1														
Belgium	13	2	3	4	3	1												
Brazil	5	1		2		1			1									
Cabo Verde	1			1														
Canada	9	1			2	1	2	1	1						1			
Chile	7	3			1		1				1							1
China	25	5	7	5	3	2	2		1									
Colombia	1				1													
Costa Rica	2	1					1											
Croatia	1					1												
Czech Republic	3			1			1			1								
Denmark	1			1														
Egypt	1	1																
Finland	2					1		1										
France	13	3	2	1	3	2	1	1										
French Polynesia	1	1																
Gabon	1											1						
Germany	40	5	9	2	4	4	6	4	3	3								
Greece	1			1											1			
Hong Kong	2		1						1									
Hungary	10	1		1	2	1	2				1		1	1				
India	4	1	2		1													
Iraq	1		1															
Ireland	1	1																
Israel	9	2		2	2					2		1						

(Continued)

Table 1. (Continued)

	Total	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010	2009	2008	2007	2006	2005	2004	2003
Italy	15	3	4	2		1		1		1					1	1		1
Japan	3	1	1		1													
Kenya	1	1																
South Korea	14		7	2	1	1	2	1										
Latvia	1			1														
Lithuania	1						1											
Malaysia	1			1														
Mexico	5	2		1	1		1											
New Zealand	2		1				1											
Norway	1		1															
Pakistan	1								1									
Panama	1		1															
Poland	10	4	1	1	1		1	1	1									
Portugal	3				1	1					1							
Romania	1								1									
Serbia	1								1									
Slovakia	10			2		1	3	1	1		1			1				
Slovenia	1					1												
South Africa	4				1	1		1				1						
Spain	28	4	3	4	5	4	2		3		1			1			1	
Sweden	3			2	1													
Switzerland	4	1	1		1	1												
The Netherlands	28	2	6	6	2	4	3		2	2			1					
Taiwan	1					1												
United Kingdom	27	2	6	2	4	7	2	2	1			1						
Turkey	1					1												
USA	91	18	14	10	11	5	5	9	1	4	3	4	1	1	1	1	2	1
2 > Country	23																	

Considering the studied area or the laboratory location.
Source: Data obtained and assessed from ISI Web of Science.

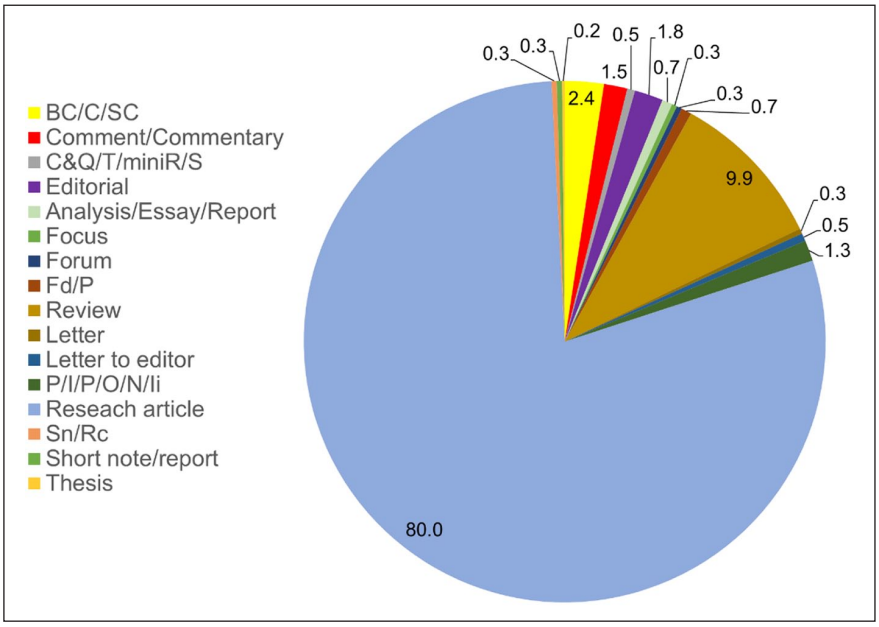


Figure 5. Percentage of publications on light pollution following specific types of formats (2003–2019).
BC/C/SC: Brief communication/Communication/Short communication; Sn/Rc: Scientific note/Research communication; C&Q/T/miniR, S: Concepts and questions/theory/reviews/syntheses; P/I/P/O/N/Ii: Preface/Introduction/Profile/Opinion/Mini review/Note/Invited ideas; Fd/P: Future directions/Perspective.
Data were obtained from ISI Web of Science.

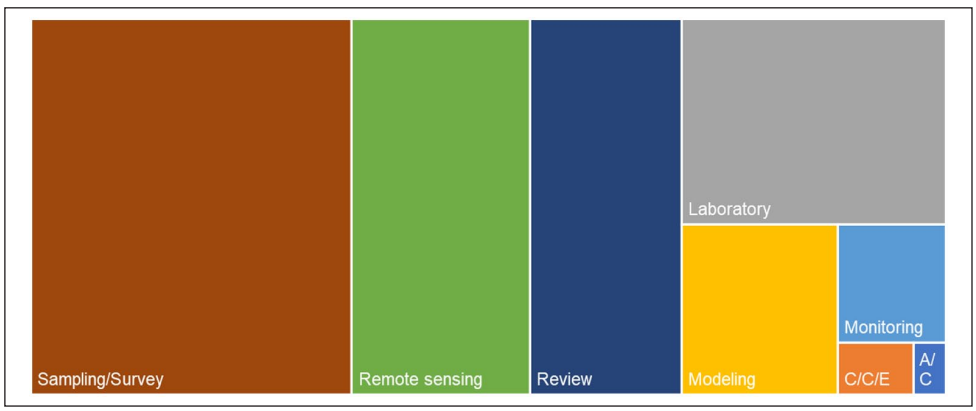


Figure 6. Percentage of publications on light pollution following specific methodologies (2003–2019).
C/C/E: Comment/Conceptualisation/Editorial; A/Cal: Assessment/Calibration.
Data were obtained from ISI Web of Science.

2017) and hand-held Sky Quality-Metres (SQM) that can be combined with Geographical Information Systems (Biggs et al., 2012; Den Outer et al., 2015; Espey and McCauley, 2014; Pravettoni et al., 2016). Also, Levin and Phinn (2016) investigated the possibility of using satellites such as the OLI optical sensor on-board the Landsat 8 to detect artificial light in the visible bands.

They found it useful only when light is emitted from very bright features, such as from urban areas and gas flares in at – sensor level measurements. These authors affirm that this type of satellite was not developed for acquiring night-time images, being less sensitive than the VIIRS Day/Night band. Data obtained from remote sensing and sensors must consider all-weather conditions. Therefore, the number of clouds should also be measured as suggested by Kotarba et al. (2019) who used surface-based (SYNOP) and satellite-based (Meteosat/SEVIRI) data. Kyba et al. (2011) demonstrated that cloud coverage could highly amplify the sky luminance depending on the closeness to the city centre. These results were significantly different from rural moonlit nights. These methods are accurate, reproducible and valid to assess large territories. However, the treatment of the data requires powerful equipment, permissions and rights that can be expensive and difficult to be acquired by researchers from low-income countries.

A total of 16.6% and 1.1% of the published papers were Reviews and C/C/E (Comment/Conceptualisation/Editorial), respectively. The number of papers highlighting light pollution as the main issue is significant. Their aims are also diverse, focussing on relevant topics such as insects and pollination (e.g. MacGregor et al., 2014), necessary strategies to be conducted against human activities such as agriculture (e.g. Habel et al., 2019), birds (e.g. Adams et al., 2019) and bird migration (e.g. Lennox et al., 2016). Also, there are some review papers related to bat behaviour, foraging, commuting, emergence, roosting and hibernation (e.g. Stone et al., 2015) and marine (Davies et al., 2014) and fluvial ecosystems (e.g. Jechow and Hölker, 2019). In this group, several manuscripts assessed the impacts of light pollution emanating from urban areas (e.g. Mockey Coureaux and Manzano, 2013) carrying out a meta-analysis and assessing own remote sensing data (e.g. Hänel et al., 2018; Zhao et al., 2019). It is important to remark some relevant concepts extracted from these articles such as the 'negative luminescence' (Gandy, 2017), 'biodiversity-sensitive cities and towns' (Ikin et al., 2015), 'light pollution and passive pleasures, and the instrumental value of beauty' (Gallaway, 2010). Also, the term 'polarised light pollution' is used as a new kind of 'ecological photopollution' (Horváth et al., 2009) or the 'ecological light pollution' (Longcore and Rich, 2004), which is referred to the alteration of natural light regimes in terrestrial and aquatic ecosystems. Finally, it is worthy to note the elevated number of review papers related to human psychiatry (e.g. Hoare et al., 2019) and wellbeing, considering topics about ageing (e.g. Parrado et al., 2019), sleep time (e.g. Mantua et al., 2019), diabetes (e.g. Afroz-Hossain et al., 2019), etc.

Laboratory (15.8%), modelling (7.7%) and monitoring (3.6%) techniques were used in combination with the above-mentioned methods. Therefore, only fewer papers have been published using these methods alone.

Impacts on different specific issues

The number of publications on light pollution considering the different impacts on specific issues, living organisms and processes are summarised in Table 2.

Natural ecosystems, animals and aquatic ecosystems. A total of 269 studies have been published on animals and light pollution. Over the last decade, there has been an explosion of investigations related to the biological impacts of light pollution such as on migration, sleep time or health (Gaston et al., 2015; Lennox et al., 2016).

Figure 7 summarises the most studied species of birds (69), insects (57) and bats (44). According to several studies conducted about birds, sleep time and behaviour did not differ under dark conditions but light pollution extremely modifies this trend depending on intensities. For example, Sun et al. (2017a) comparing blue and great tits demonstrated more pronounced effects and more aspects of sleep were affected for the latter. They fell asleep later, woke up and exited the nest

Table 2. Number of publications on light pollution considering its impacts on the different elements from 2003 to 2019.

Impacts on	Total		Specific impacts
Advertising signs/Luminaires	2		
Animals	267	Bats	44
		Lemurs/roe/mice/mouse/squirrel/ hamster/vole	13
		Birds	70
		Crayfishes	3
		General studies	11
		Lizards/geckos/salamander	4
		Primates, wallabies	3
		Toads and frogs	9
		Turtles	19
		Mollusca/crustacean	5
		Aquatic invertebrates	5
		Fishes	23
		Gastropod	3
		Insects	57
Aquatic ecosystems	21	Benthic	2
		Corals	2
		Periphyton/plankton	3
		Aquatic ecosystem/freshwater (general)	14
Ecological communities	22		
General (skyglow, night sky brightness, night time, etc.)	125		
Astronomy/tourism	6		
Engineering	29	Engineering developments	6
		Energy consumption	1
		Domestic lights	1
		Lamps/LEDs	21
Human health	75	Atherosclerosis, hypersensitivity, diabetes, disease diffusion	4
		Body mass	4
		Cancer	12
		Cardio	1
		Circadian rhythm	4
		Mental disorders (depression, Alzheimer, Parkinson)	12
		Eyes	6
		General aspects	10
		Skin	3
		Sleeping time	10
		Human health (the use of screens)	1
Sociological aspects	10	Human perception	7
		Human health/landscape evaluation	1
		Ethical concept	1
		Economy	1

(Continued)

Table 2. (Continued)

Impacts on	Total	Specific impacts
Land use changes	2	
Plants	9	
Policies/protected areas	3	
Urban/rural areas/roads	101	

Source: Data obtained and assessed from ISI Web of Science.

boxes earlier, and the total sleep amount and sleep percentage were reduced. These findings confirmed one specific species cannot be used as a generalisation, even if they are closely related. However, these changes could also affect the whole community. Poot et al. (2008) obtained clear results about nocturnally migrating birds. They found that these birds were disoriented and attracted by red and white lights. However, this situation changed with blue and green lights, opening a new challenge to develop bird-friendly artificial lights by changing colours. Dominoni and Partecke (2015) found that artificial light at night can modify biological rhythms in avian urban-dwellers. This perception can even change if the sex of the birds is considered (Dominoni et al., 2013). Other authors found contrary results. For example, Silva et al. (2017) did not observe any signal that birds advance the onset of foraging in gardens with more artificial lights nearby contrary to their first hypothesis. They considered that it could be due to a response to differences in food availability and predation risk in areas better illuminated. Also, Clewley et al. (2016) observed that artificial light could not be the main driving factor concerning the timing of foraging behaviour during winter. Raap et al. (2017) highlighted no significant results for seasonal and light intensity-dependent effects of artificial lights and recommend further research using lower light intensities. This study contradicted earlier results where light pollution showed a clear disruptive impact on the sleep of wild animals.

With insects, no definitive results have been published. For example, two different groups debated on the possibilities to create demographic traps for nocturnal insects by intense light pollution (Firebaugh and Haynes, 2019; Lewis and Owens, 2019). Aerial insects are being highly affected by light pollution. Wakefield et al. (2018) compared insect attraction to high-pressure sodium and metal halide and LED lights on the streets. They demonstrated that white LEDs do not attract more insects than the others, highlighting the necessity to use these lights to reduce disturbances across multiple insect taxa. Recently, Larsson et al. (2020) demonstrated similar behaviour as the above-mentioned for birds, but for *Trichoptera*, whose attraction to light varied among species, females are more influenced than males. These authors registered an atypical flight activity when the lamps were turned on at night. Yuen and Bonebrake (2017) observed that artificial lights at night could reduce prey interception for spiders too, which makes simple predictions of ecological light pollution's disruption of species interactions difficult. Both aerial and terrestrial insects are being affected. In their recent review papers, Habel et al. (2019) and Fox (2013) estimated a clear reduction of terrestrial insects among others, adding light pollution to other supplementary and adversely synergistic factors such as climate change, urbanisation and associated processes (e.g. traffic volume or air/water pollution).

Bats are nocturnal animals, and, thus, they are one of the most vulnerable species affected by light pollution (e.g. Lacoeuilhe et al., 2014; Musila et al., 2019; Straka et al., 2019). Rydell et al. (2017) highlighted that artificial lights of churches, and other historical buildings for aesthetic purposes, are some of the major problematic places for bats. However, this environmental issue is generally not considered in environmental impact assessments by public agencies. They proposed

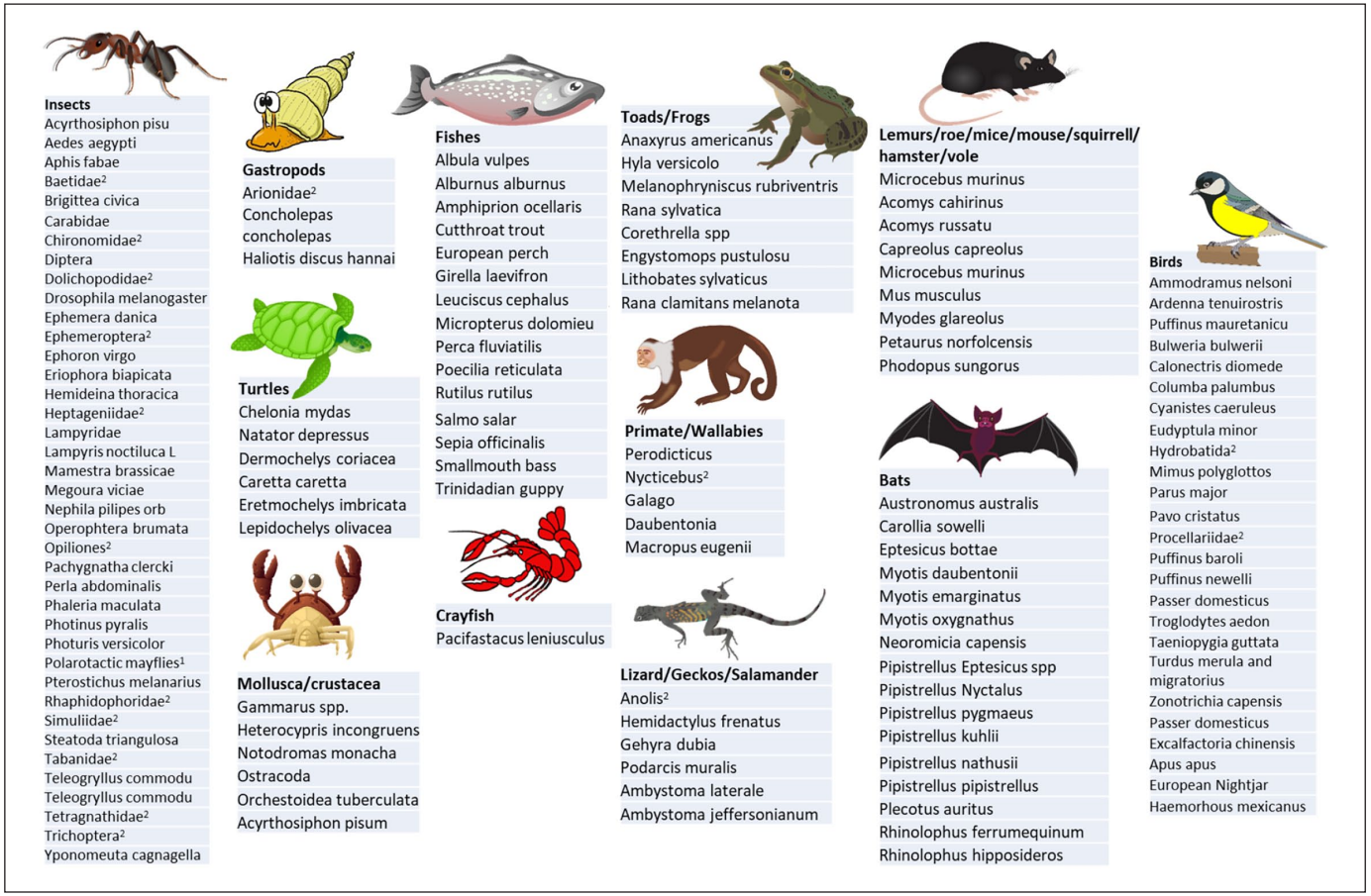


Figure 7. List of species studied for possible impacts of light pollution (2003–2019).
1: common names; 2: family or order.
Animal's pictures are obtained from Pixabay (https://pixabay.com/), which offers free images.
Data were sourced from ISI Web of Science.

the installation of flood-lights on these structures. To reduce the impacts of light pollution on bats behaviour and foraging, Stone et al. (2015) proposed the avoidance of outdated, poorly installed and/or maintained lights, modification of the duration and timing of lighting regimes, its intensity and the types. To date, the impacts of light pollution on bats and hibernation, fitness costs or long-term implications have not been addressed. Azam et al. (2018) suggested separating streetlights from ecological corridors (≈50 m) without vertical light trespass.

Also, it is relevant to highlight the number of articles aimed to assess animals in aquatic environments such as fishes (23), diverse aquatic ecosystems (21), turtles (19) or toads and frogs (9). Depledge et al. (2010) remark in their editorial that surprisingly light is one particular pollutant permitted to flood into the aquatic ecosystems and it is not foreseen to be addressed in the near future. Increasing illumination of the coastal zones, reefs, rivers, estuaries, deltas, gulfs and nearshore areas is threatening aquatic animals all over the world (e.g. McMahon et al., 2017; Nobes et al., 2008; Tamir et al., 2017; Zapata et al., 2019), even across deep profiles (Bolton et al., 2017). Davies et al. (2016) stated that it is imperative to improve our understanding of the ecological impacts of light pollution on marine ecosystems and they proposed the allocation of 'Marine Dark Sky Park'.

Other studies assessing the impacts of light pollution focussed on lemurs, roes, mice, squirrels, hamsters, voles, lizards, geckos, salamanders, some primates or wallabies. In this group, it is relevant to assess the results to evaluate potential human responses and the elevated percentage of studies conducted under laboratory conditions.

Human health and sociological aspects. In their review paper, Davies and Smyth (2018) indicated that human health and human-environment interrelationships must be considered as key questions for light pollution research in the 21st century, among other topics such as diversity and sensibility of biological responses, solutions or the expansion of LED lights. Dominoni et al. (2016) published an opinion piece that affirmed the scarcity of information on light pollution and the underlying physiological mechanisms affecting metabolic syndromes, cancer and reduced immunity of humans. We found a total of 77 articles published that shed new insights on this topic.

Without a doubt, the most well-known negative impacts of artificial lights are on circadian rhythms and sleep disorders (Cho et al., 2013; Touitou, 2015). However, some studies verified that practical effect sizes could be smaller than expected and more associated with idiosyncratic issues (Patel, 2019). Twelve papers aimed to investigate the impacts of light pollution on mental disorders such as depression, Alzheimer and Parkinson (e.g. Kim et al., 2018). Moreover, some authors observed that elevated exposure under artificial lights at night could be linked to the prevalence of depressive symptoms and suicidal behaviour. We found two types of studies confirming these effects depending on the method. The first, type of studies used rodents while the second used representative human samples (Bedrosian and Nelson, 2013). For example, Bedrosian et al. (2011) and Fonken and Nelson (2013) found that light pollution can contribute to changes in the mood of hamsters and mice, which could be vital for evaluating and developing solutions for depressive disorders. On the other hand, Min and Min (2018) developed cross-sectional research using a representative sample of Korean inhabitants and found that light pollution could be considered as an 'environmental contributor' to mental health problems.

Ten articles are related to the occurrence of cancers. Giraudeau et al. (2018) highlighted that humans can be defined as an 'oncogenic species' because they modify the environmental conditions causing cancer in other wild populations, or even, in other humans. In the study conducted by Xu and Gao (2017), an increase in artificial lights has a positive correlation with health index for inhabitants older than 60 ages (for 2011), cancer rate (for 2009) and surface temperature in Beijing (for 2012).

We observed that breast cancer is the main studied type of cancer studied. It is related to the nocturnal suppression of melatonin production in the pineal gland under a high artificial light exposure, which some authors hypothesise that could be reduced by avoiding light at night (Keshet-Sitton et al., 2016). The cancer risks are associated with both women men, and increasing the duration of night work is not the key factor (Parent et al., 2012).

It is worth highlighting other specific investigations focussed on the possible effects related to atherosclerosis (Obayashi et al., 2019), body mass (Fonken et al., 2010), ocular issues such as fatigue (Park, 2018), retina degeneration (Benedetto and Contin, 2019), skin deterioration (Parrado et al., 2019) or even diabetes (Afroz-Hossain et al., 2019).

Despite the elevated number of studies confirming the negative impacts on human health, there is ongoing debate concerning the necessity to reduce or eliminate lights but lights in urban help to reduce crime, improves transportation safety and minimises concealment, aesthetic pleasing and helps with astronomical observations (e.g. Chepesiuk, 2009; Jones, 2018; Kim et al., 2015). Nevertheless, Schulte-Römer et al. (2019) prefer distinguishing between 'lighting professionals' having lighting design or/and industry background, and 'light pollution experts' who specialised in astronomy and environmental science issues. Fortunately, these authors agree on the issue definition and problem-solving advances facing light pollution, but they disagree on some other questions such as potential difficulties to mitigate light pollution impacts and related governance challenges. Also, some other studies show how ordinary inhabitants agree with some questions related to artificial lights. For example, Haans and de Kort (2012) surveying stationery and walking participants, confirmed that urban inhabitants prefer to illuminate their close surroundings instead of the roads. In Finland, 2053 respondents considered light pollution as an issue for outdoor recreation and relaxation and should be considered by policymakers and land planners (Lyytimäki and Rinne, 2013). These ambiguities manifest the necessity to consider light pollution with hindsight to their moral implications and ethical views (Stone, 2017).

Light pollution in general terms. We observe that 125 articles aimed to investigate general or specific aspects related to skyglow, night-time and night sky brightness.

Urban measurements differed from those in suburban territories, which could demonstrate a mixed-effect model with different results at a regional scale (Fu et al., 2018; Kuechly et al., 2012; Ściężor et al., 2012). This makes astronomical observations difficult as the lighting sources are far away from the Earth (Barducci et al., 2003; Plauchu-Frayn et al., 2017). Therefore, light pollution indexes (e.g. Saraiji and Oommen, 2012; Votsi et al., 2017) and theoretical/mathematical models (e.g. Kocifaj, 2008, 2011) are useful tools considering the size, shape, spatial distribution, radiative pattern and spectral characteristics from the light sources, atmospheric conditions and orography. These new advances are helping to develop new maps (e.g. as the world atlas of artificial sky luminance) and quantifications for the most affected areas, although there is still some weakness yet to be resolved (Falchi et al., 2016; Kyba et al., 2013, 2015), or other large scale assessments (Bennie et al., 2015; Cauwels et al., 2014). Undoubtedly, this is possible because of the new advances and development of powerful techniques related to remote sensing, geographical information systems and statistical software. We hypothesise that this research line will be one of the most exploited in the future by the scientific community. However, the negative impacts of light pollution on human and ecosystem health could be relegated to the second level of importance because of the increasing and sudden apparition of other issues such as the COVID-19, which is attracting the majority of efforts and economic supports by institutions and research groups (Pai, 2020; Woolston, 2020).

Economy, urban and rural areas, and land-use change. It is well-known that light pollution is an environmental issue with clear economic implications (Gallaway et al., 2010; Olsen et al., 2014). We

are experiencing never seen growing light use but a drastic reduction remains an ongoing challenge without strong legislation, funding and guidelines/policies for stakeholders and inhabitants (Feder, 2005; Stanley et al., 2015). To date, research on the social-economic effects of light pollution has been neglected. One notable example would be the publication by Ngarambe et al. (2018) using the Kuznets curve for variable light pollution and land prices. This method shows the interconnected relationship among variables and has been successfully using for other environmental issues (Mosconi et al., 2020). In this study, however, their results did not show an inverted-U relationship between the two above-mentioned variables.

A total of 29 articles highlight new engineering solutions aimed to reduce light pollution that focus on LEDs (light-emitting diode) and other lamps of a new generation (e.g. Narendran et al., 2016; Quintana-Barcia et al., 2018; Sun et al., 2017b). It is well known that outdoor streetlights are among the most expensive costs for municipalities, but there is limited research conducted to assess their efficiency. Mockey Coureaux and Manzano (2013) indicated that one of the major causes of black-outs could be the accumulation of dirt on the luminaires (lamps and lighting fixtures), which directly influence depreciation. Light pollution from indoor artificial lighting in office buildings are usually not considered for human wellbeing or large scale assessments (Du et al., 2018).

It was estimated that LED could reach 70% of the outdoor and residential lighting markets by 2020, but there is a relevant debate about the impacts on wildlife, animal welfare and disease transmission (Wakefield et al., 2016). In fact, Pawson and Bader (2014) showed that on average traps equipped with LEDs were able to catch 48% more insects than those with HPS (high-pressure sodium vapour lamps), but there was no evidence that colour temperature changes of white LEDs can reduce the ecological impacts. These authors asserted that it is vital to organise collaborative investigations between scientists and engineers to avoid or at least minimise their potential negative impacts. However, some positive results were obtained in relation to human wellbeing and the economy. By estimating the local externalities of investments in LEDs in most populated US counties, Jones (2018) confirmed the significant positive impacts on human well-being, with some variations among respondent demographic types. The most important economic result was related to the per capita monthly household income which amounted to $477 (6.9%).

It is interesting to remark that >100 articles were published on the impacts of light pollution within urban, suburban, peri-urban and rural areas and some possible modifications in parallel with land use cover changes. This topic is highly discussed in the recent literature. Hopkins et al. (2018) carefully insisted that light pollution is a ubiquitous feature of urbanisation that change over time and shows an elevated potential to exert strong selection pressures on living organisms, especially, in urban environments. In this regard, Gandy (2017) highlighted that investigation of light pollution mentioning some approaches such as ‘urban atmospheres’, ‘urban ecology’ or ‘smart cities’ are not able to understand the complete difficulty of the politics of light under earlier modernity. This author affirms that the ‘rise of light-emitting diodes (LEDs) or the capitalist transformation of the nocturnal realm’ are usually not well understood. Therefore, we would want to consider new articles that are helping to improve these valid concepts or approaches and its lacks, for example, the modelling research conducted by Ziou and Kerouh (2018) for smart cities, or by Dominoni and Nelson (2018) highlighting the relevance of integrative studies across taxa, biological functions and scientific disciplines from the urban ecology approach. In this line, we strongly consider excellent contributions which can afford: (i) the effects of construction region environments on night-time brightness considering the regional economic development levels (Wang et al., 2019); (ii) the definition of the city boundaries with dramatic impacts on the scaling dynamic of the night-time light (Operti et al., 2018) and (iii) the combination of technical studies (e.g. using remote sensing data or surveys) and considerations for politicians, decision- and law-makers or governmental agencies (Schuler et al., 2018).

Conclusions and main remarks

In this review article, we have assessed the literature on the impacts of light pollution on the ecological and human spheres. This is a complement to other reviews on light pollution and updated to 2020. We evaluated >600 scientific documents and confirmed the elevated and diversified nature of research on light pollution considering the broad range of methods used, elements affected and scientific journals published in. However, light pollution is highly affected by external decisions made by stakeholders, ordinary inhabitants, policymakers and politicians from municipalities, regions and countries. These decisions are clearly affecting humans and the natural ecosystem health, but the experts and inhabitants have divergent perspectives, making light pollution a multifaceted term and difficult to be faced, mitigated and not holistically understood. We strongly agree with Clanton (2014) who formulated simple rules to help drastically reduce light pollution, namely: (i) using lights only when necessary; (ii) employing the lowest illuminance as possible and (iii) minimising the light sheds skywards (particularly any close to the horizontal). Also, Gaston et al. (2012) highlighted the necessity to reduce the duration trespass of lighting into territories that are not intended to be lit (including the night sky) and to modify the intensity and spectral composition of lighting. We also add the importance of teaching humankind and promoting engineering research to develop more friendly, aesthetically pleasing and sustainable lighting, taking humans and the natural ecosystems into consideration. This will help to reduce the economic wastes and improve the nighttime, pleasure to observe the sky, nocturnal landscapes and astronomical observations.

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ORCID iD

Jesús Rodrigo-Comino https://orcid.org/0000-0002-4823-0871

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